

Rethinking Social Computing Research in the Age of AI: Agency, Collaboration, and Governance

Yun Huang
School of Information Sciences
University of Illinois at
Urbana-Champaign
Champaign, Illinois, USA
yunhuang@illinois.edu

Noshir Contractor
Industrial Engineering &
Management Sciences
Northwestern University
Evanston, Illinois, USA
nosh@northwestern.edu

Tai-Quan Peng
Department of Communication
Michigan State University
East Lansing, Michigan, USA
pengtaiq@msu.edu

Dashun Wang
Kellogg School of Management
Kellogg School of Management
Evanston, Illinois, USA
dashun.wang@kellogg.northwestern.edu

Cuihua Shen
UC Davis
Davis, California, USA
cuishen@ucdavis.edu

Manling Li
Computer Science
Northwestern University
Evanston, Illinois, USA
manling.li@northwestern.edu

Yiren Liu
Informatics
University of Illinois
Urbana-Champaign
Champaign, Illinois, USA
yirenli2@illinois.edu

Hangyue Zhang
School of Information Sciences
University of Illinois
Urbana-Champaign
Champaign, Illinois, USA
hz85@illinois.edu

Sangho Suh
The Allen Institute for AI
Seattle, Washington, USA
sanghos@allenai.org

Abstract

Recent advances in large language models have intensified public debate about the role of AI in supporting research pipelines, from hypothesis generation to experimental planning, execution, communication, and evaluation. While pattern recognition and automation improve efficiency, scholars are also raising deeper concerns about agency, creativity, intellectual ownership, and the potential erosion of human judgment in scientific work. This Special Interest Group (SIG) aims to create an open space for discussing emerging opportunities and challenges in AI-supported social computing research. We will discuss AI use across diverse research contexts, including individual inquiry, team formation and collaboration, peer review of papers and proposals, interdisciplinary knowledge production, and governance. We will also address the responsible design of AI systems that support deeper insights with greater confidence, promote new team collaboration and trust, and intentionally introduce productive frictions that help cultivate critical and creative thinking. By bringing together scholars from HCI, AI, the social sciences, and related fields, this SIG will develop a shared blueprint for an AI-empowered social computing research agenda. More importantly, it will create a space for researchers to revisit their values, reaffirm human-centered commitments, and cultivate resilience in an ever-evolving AI era.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing systems and tools.**

Keywords

AI-supported research, social computing, large language models, human-centered AI

ACM Reference Format:

Yun Huang, Noshir Contractor, Tai-Quan Peng, Dashun Wang, Cuihua Shen, Manling Li, Yiren Liu, Hangyue Zhang, and Sangho Suh. 2026. Rethinking Social Computing Research in the Age of AI: Agency, Collaboration, and Governance. In *Companion of the Computer-Supported Cooperative Work and Social Computing (CSCW Companion '26)*, October 10–14, 2026, Salt Lake City, UT, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3785651.3831580>

1 Introduction

Large language models have demonstrated strong performance in programming and analytical tasks, fundamentally changing how the industry produces software. In the scholarly community, ChatGPT, Gemini, and Asta have offered “deep research” features [2], which facilitate the synthesis of related work, the generation of research questions, and the simulation of experimental results.

A recent comprehensive survey also positions “Agentic Science” as a new paradigm in which AI systems autonomously generate hypotheses, design experiments, and analyze results [21], while others envision “co-evolving ecosystems” of human and AI scientists [17]. A landmark study with over 100 NLP researchers found that LLM-generated research ideas were rated as more novel than human-generated ones, though slightly weaker on feasibility [18]. Meanwhile, a large-scale survey of researchers revealed widespread



This work is licensed under a Creative Commons Attribution 4.0 International License.
CSCW Companion '26, Salt Lake City, UT, USA
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2378-0/26/10
<https://doi.org/10.1145/3785651.3831580>

but uneven adoption of LLMs, with significant concerns about reliability, bias, and the erosion of core research skills [8]. These developments underscore a fundamental tension: as AI becomes more capable of performing research tasks autonomously, the human role in scientific inquiry and teamwork must be deliberately preserved and redefined.

Unlike traditional STEM fields, social computing is highly interdisciplinary [19]. When conducting social computing systems research, scholars often need to understand technological abilities and constraints, draw theories from psychology, social sciences, and organizational research, design mixed methods, and develop iterative design and empirical evaluations. Recently, HCI scholars have developed LLM-based systems to support creative research ideation rather than task automation alone [9–11, 14, 23]. For example, in our own research, we found that the most valuable contributions of AI were not factual correctness or speed, but its ability to offer unfamiliar perspectives and provoke reflection [23]; AI delays can prompt a stronger sense of agency among researchers when exploring research gaps [9]; and debates among multiple AI agents encouraged researchers to compare, contrast, and critique alternative framings of the same problem [10].

Across these iterations, we observed a consistent pattern: the most rewarding moments occurred when AI created space for human deep work [10, 11]. These empirical findings challenge the prevailing assumption that the primary goal of AI-supported research is to maximize efficiency.

Indeed, as Ottino and Uzzi argue in a recent Science editorial [13], science advances not only through efficiency and optimization, but also through redundancy, divergent exploration, failed attempts, and partially independent lines of inquiry. In other words, some inefficiencies are not flaws in the system but essential features that preserve epistemic diversity, creativity, critical evaluation, and conceptual breakthroughs.

2 SIG Goals

The goal of this SIG is to create an open space for discussing emerging opportunities and challenges in AI support for social computing research. This SIG is inspired by the 2026 Lambert ANN SONIC NICO Workshop, co-directed by Dr. Noshir Contractor and Dr. Yingdan Lu. More than 40 researchers from diverse backgrounds were invited to discuss human-centered computing research in the age of AI. Several of the proposed SIG topics are informed by the workshop discussions. These conversations are also expanding through broader interdisciplinary venues, including the Stanford CASBS Summer Institute on AI Methods in the Social Sciences, connecting scholars across HCI, communication, organizational research, AI, and the social sciences.

Specifically, the following themes of AI support and questions will be discussed:

- **Knowledge building and research inquiry:** While AI can enhance cognitive capacity, prolonged reliance risks eroding the very skills researchers depend on, including: critical analysis, methodological judgment, and creative synthesis. These concerns are particularly acute for early-career researchers, for whom foundational skill development may

be disrupted by premature AI delegation. What does ethical use of AI agents mean to support short- and long-term knowledge acquisition or research inquiry? How can AI support researchers in building lasting understanding rather than merely producing quick answers? What designs and interaction modalities encourage reflection, skepticism, and intellectual ownership? What forms of AI literacy are essential for researchers to conduct independent research inquiry?

- **New team formation and small/large team collaboration:** The emerging field of “AI-teaming” [16] examines trust, coordination, and shared mental models when AI agents participate in human teams dynamics that are especially complex in interdisciplinary research settings where disciplinary power asymmetries already exist. As AI begins to participate directly in meetings and collaborative workflows, note-taking bots, real-time summarizers, and emerging facilitative agents are altering core meeting dynamics [1]. What are the appropriate boundaries for using AI to facilitate team formation, communication, and research development? How can AI help researchers discover complementary expertise, negotiate shared goals, and coordinate collaboration across disciplinary boundaries? How do different forms of AI-mediated communication affect trust, participation, and power dynamics across varying scales of collaboration? When might AI recommendations reinforce existing academic hierarchies or biases?
- **Community building through peer reviews:** A substantial fraction of peer reviews at major AI conferences now contain AI-modified content [7]. While a large-scale randomized study at ICLR 2025 found that LLM feedback can improve review quality by prompting reviewers to address overlooked issues [20], other studies reveal that fully automated reviewers fail to detect faulty reasoning [3]. Besides evaluating papers, peer review also supports community building and prompts mutual respect. Which parts of the review process can be responsibly supported by AI, e.g., checking clarity, identifying missing literature, identifying methodological flaws, or comparing reviewer comments? Which human tasks should remain non-replaceable, e.g., taking responsibility for recommendations? How might AI affect community building and accountability of scholarly works?
- **Interdisciplinary research development and evaluation:** Early benchmarking efforts such as SoundnessBench [6] attempt to evaluate whether AI agents can distinguish sound research ideas from flawed ones. Should we teach AI systems to recognize, support, or challenge “good research taste”? If so, how should research taste be defined across disciplines, methods, and communities? What could go wrong if AI systems begin to shape what counts as novel, rigorous, fundable, or publishable research?
- **Authorship, ownership, and governance:** A scoping review of generative AI in scholarly writing [15] documents the rapid shift from problem-framing to explicit policy development across journals and conferences, yet significant gaps remain particularly around crediting AI contributions to ideation and analysis, as distinct from text generation.

Table 1: Schedule for the 90-minute CSCW SIG session.

Time	Event
0–5 min	Welcome & introduction
5–40 min	Lightning talks & short panel
40–45 min	Transition & preparation for group work
45–80 min	Theme-based group discussions
80–90 min	Report-backs, summarization, & closing remarks

How should researchers disclose, credit, and take responsibility for AI-supported research ideation, development, and evaluation? Who owns ideas, text, data interpretations, or designs? What assumptions are challenged? What new norms might be expected?

Across these scenarios, we seek to explore the evolving human-AI interaction in scholarly work [4, 5, 12, 14, 21, 22]. What potential risks or mistakes are part of the growth pain, and which are not? How do we distinguish harmful inefficiency from productive inefficiency? Which forms of redundancy, disagreement, or slower reflection are essential for creativity and trustworthy science? How can researchers cultivate resilience in an increasingly AI-mediated research environment? What are our non-negotiable value propositions?

3 Session Plan and Schedule

Before the SIG session, we will create a GitHub repository and a website for attendees to share materials, propose topics, and begin networking prior to the conference. The organizing team brings together complementary expertise spanning human-computer interaction (Huang, Liu, and Suh), computational social science (Contractor and Peng), science of science (Wang), natural language processing (Li), and AI governance (Shen). We will invite participants through social media promotion, professional mailing lists, and our existing research networks.

At the conference, we plan a 90-minute session, with a tentative schedule detailed in Table 1.

Lightning Talks & Short Panel. Experts from academia and industry will be invited to give 5-minute lightning talks introducing the key themes of AI-supported research. This will be followed by a short panel discussion, including Q&A among the moderator, panelists, and the audience, to provoke initial thoughts and debates.

Theme-based Group Discussions. We will host a 35-minute group activity using a “birds of a feather” format. Participants can choose to join one of the predefined themes (e.g., individual inquiry, team formation, peer review, authorship). Each group will brainstorm the proposed topics and at least one organizer will join each group to facilitate the discussion.

3.1 Expected Attendees and Post-SIG Engagement

We expect these topics to attract a similarly diverse community, including participants with varying levels of familiarity, expertise, and seniority in AI and social computing. Attendees will include academic and industry researchers, students, and practitioners who

are broadly interested in how AI is reshaping social computing research. Participants are expected to come from fields such as human-computer interaction, artificial intelligence, psychology, the social sciences, and organizational behavior. We anticipate a group of 30–50 participants for in-depth discussion and will adjust the structure as needed to accommodate a larger group. We will share the SIG’s key discussions, related documents, and findings through a GitHub repository to reach a broader audience and sustain ongoing dialogue beyond the session.

Acknowledgments

This material is based upon work supported by the Institute of Education Sciences, U.S. Department of Education, through Grant 2229873 (NSF). Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

References

- [1] Joseph A Allen. 2026. From human gatherings to human-agent meetings: Rethinking small group and team collaboration in the age of AI. *Small Group Research* (2026), 10464964261430677.
- [2] Allen Institute for AI. 2026. Asta: Scholarly Research Assistant. <https://asta.allen.ai/>. Accessed: 2026-06-22.
- [3] Nils Dycke and Iryna Gurevych. 2025. Automatic Reviewers Fail to Detect Faulty Reasoning in Research Papers: A New Counterfactual Evaluation Framework. *Transactions of the Association for Computational Linguistics* (2025).
- [4] Edona Elshan, Dominik Siemon, Triparna de Vreede, Gert-Jan de Vreede, Sarah Oeste-Reiss, and Philipp Alexander Ebel. 2022. Requirements for AI-based Teammates: A Qualitative Inquiry in the Context of Creative Workshops. In *Hawaii International Conference on System Sciences*. <https://api.semanticscholar.org/CorpusID:248672379>
- [5] Alireza Ghafarollahi and Markus J. Buehler. 2024. SciAgents: Automating Scientific Discovery Through Bioinspired Multi-Agent Intelligent Graph Reasoning. *Advanced Materials (Deerfield Beach, Fla.)* 37 (2024). <https://api.semanticscholar.org/CorpusID:274858957>
- [6] Sy-Tuyen Ho, Minghui Liu, and H. Nghiem. 2026. SoundnessBench: Can Your AI Scientist Really Tell Good Research Ideas from Bad Ones? (2026).
- [7] Weixin Liang, Zachary Izzo, and Yaohui Zhang. 2024. Monitoring AI-Modified Content at Scale: A Case Study on the Impact of ChatGPT on AI Conference Peer Reviews. In *Proceedings of the International Conference on Machine Learning (ICML)*.
- [8] Zhehui Liao, Maria Antoniak, Inyoung Cheong, Evie Yu-Yen Cheng, Ai-Heng Lee, Kyle Lo, Joseph Chee Chang, and Amy X Zhang. 2024. Lms as research tools: A large scale survey of researchers’ usage and perceptions. *arXiv preprint arXiv:2411.05025* (2024).
- [9] Yiren Liu, Si Chen, Haocong Cheng, Mengxia Yu, Xiao Ran, Andrew Mo, Yiliu Tang, and Yun Huang. 2024. How AI Processing Delays Foster Creativity: Exploring Research Question Co-Creation with an LLM-based Agent. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI ’24). Association for Computing Machinery, New York, NY, USA, Article 17, 25 pages. doi:10.1145/3613904.3642698
- [10] Yiren Liu, Viraj Shah, Sangho Suh, Pao Siangliulue, Tal August, and Yun Huang. 2026. Perspectra: Choosing Your Experts Enhances Critical Thinking in Multi-Agent Research Ideation. *CHI* (2026).
- [11] Yiren Liu, Pranav Sharma, Mehul Oswal, Haijun Xia, and Yun Huang. 2025. PersonaFlow: Designing LLM-Simulated Expert Perspectives for Enhanced Research Ideation. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*. 506–534. LLM-based multi-persona system for multi-stakeholder ideation.
- [12] Amogh Mannekote. 2023. Towards a Neural Era in Dialogue Management for Collaboration: A Literature Survey. *ArXiv abs/2307.09021* (2023). <https://api.semanticscholar.org/CorpusID:259950719>
- [13] Julio M. Ottino and Brian Uzzi. 2026. Progression without progress. *Science* (2026). doi:10.1126/science.aeh8945
- [14] Kevin Pu, KJ Kevin Feng, Tovi Grossman, Tom Hope, Bhavana Dalvi Mishra, Matt Latzke, Jonathan Bragg, Joseph Chee Chang, and Pao Siangliulue. 2025. Ideasynth: Iterative research idea development through evolving and composing idea facets with literature-grounded feedback. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–31.

- [15] Lilia Raitskaya and Elena Tikhonova. 2025. The 2025 Landscape of Generative AI in Scholarly Writing and Publishing: A Scoping Review of Uses and Ethical Approaches. *Journal of Language and Education* (2025).
- [16] Jan B Schmutz, Neal Outland, Sophie Kerstan, Eleni Georganta, and Anna-Sophie Ulfert. 2024. AI-teaming: Redefining collaboration in the digital era. *Current Opinion in Psychology* 58 (2024), 101837.
- [17] Chenyang Shao, Dehao Huang, Yu Li, Keyu Zhao, Weiwan Lin, Yining Zhang, Qingbin Zeng, Zhiyu Chen, Tianxing Li, Yifei Huang, et al. 2025. OmniScientist: Toward a Co-evolving Ecosystem of Human and AI Scientists. *arXiv preprint arXiv:2511.16931* (2025).
- [18] Chenglei Si, Diyi Yang, and Tatsunori Hashimoto. 2025. Can llms generate novel research ideas? a large-scale human study with 100+ nlp researchers. 2025 (2025), 94003–94092.
- [19] Constantine Stephanidis, Gavriel Salvendy, Margherita Antona, Jessie YC Chen, Jianming Dong, Vincent G Duffy, Xiaowen Fang, Cali Fidopiastis, Gino Fragomeni, Limin Paul Fu, et al. 2019. Seven HCI grand challenges. *International Journal of Human-Computer Interaction* 35, 14 (2019), 1229–1269.
- [20] Nitya Thakkar, Mert Yuksekgonul, and J. Silberg. 2025. Can LLM feedback enhance review quality? A randomized study of 20K reviews at ICLR 2025. *arXiv preprint* (2025).
- [21] Jiaqi Wei, Yuejin Yang, Xiang Zhang, Yuhan Chen, Xiang Zhuang, Zhangyang Gao, Dongzhan Zhou, Guangshuai Wang, Zhiqiang Gao, Juntao Cao, et al. 2025. From ai for science to agentic science: A survey on autonomous scientific discovery. *arXiv preprint arXiv:2508.14111* (2025).
- [22] Pengyu Zhao, Zijian Jin, and Ning Cheng. 2023. An In-depth Survey of Large Language Model-based Artificial Intelligence Agents. *ArXiv abs/2309.14365* (2023). <https://api.semanticscholar.org/CorpusID:262824840>
- [23] Qingxiao Zheng, Minrui Chen, Pranav Sharma, Yiliu Tang, Mehul Oswal, Yiren Liu, and Yun Huang. 2025. EvAlignUX: Advancing UX Evaluation through LLM-Supported Metrics Exploration. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. 1–25. LLM-supported system for evaluation planning and risk-aware UX design.